

Member 3

Name: MD Fahim Rana

ID: 24524203101

Topic: Decision Table Testing (DTT) + State Transition Testing (STT) + System Testing

RAG System Testing

Designed Test Case

Verify Retrieval and Response Generation Decision Logic

Field	Details
Test Case ID	TC-RAG-DTT-01
Test Case Title	Verify Retrieval and Response Generation Decision Logic
Test Scenario	Verify that the RAG system performs the correct action for combinations of query validity, document availability, and LLM availability.
Preconditions	1. RAG system is online. 2. Retrieval and LLM services are configured. 3. Decision logic is implemented.
Test Steps	1. Execute the test with each condition combination. 2. Observe the system action for each combination. 3. Compare the action with the decision table.
Input Data	- Y/Y/Y: Valid Query, Document Found, LLM Available - N/Y/Y: Invalid Query, Document Found, LLM Available - Y/N/Y: Valid Query, Document Not Found, LLM Available - Y/Y/N: Valid Query, Document Found, LLM Unavailable
Expected Result	Y/Y/Y → Generate Answer; N/Y/Y → Reject Query; Y/N/Y → No Information Found; Y/Y/N → Service Error.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Decision Table Testing (DTT)

Designed Test Case

Verify Query Processing State Transitions

Field	Details
Test Case ID	TC-RAG-STT-01
Test Case Title	Verify Query Processing State Transitions
Test Scenario	Verify that a submitted query moves through the expected RAG processing states without invalid transitions.
Preconditions	1. RAG system is online. 2. Retrieval and generation services are available. 3. State/status handling is enabled.
Test Steps	1. Start from Idle. 2. Submit a valid query. 3. Observe retrieval. 4. Observe answer generation. 5. Confirm completion.
Input Data	- Idle → Query Submitted → Document Retrieved → Answer Generated → Completed
Expected Result	System follows the expected state sequence and reaches Completed without an invalid transition.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	State Transition Testing (STT)

Designed Test Case

Verify Complete End-to-End RAG Workflow

Field	Details
Test Case ID	TC-RAG-SYS-01
Test Case Title	Verify Complete End-to-End RAG Workflow
Test Scenario	Verify complete system functionality from user query through retrieval, generation, and final response.
Preconditions	1. RAG system is online. 2. Knowledge base is indexed. 3. LLM is active. 4. Citations are enabled.
Test Steps	1. Open the query interface. 2. Submit "Explain vector embeddings." 3. Verify relevant documents are retrieved. 4. Verify an answer is generated. 5. Verify supporting citations are displayed.
Input Data	- Query: Explain vector embeddings.
Expected Result	Relevant documents are retrieved, a grounded answer is generated, and supporting citations are displayed correctly.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Full System Testing